

Practice Questions: Python Lists

A. Basic List Creation & Access

1. Create a list of **5 fruits** and print the **first** and **last** fruit. *Hint: Use index 0 and length minus one.*
2. Create a list containing **integer, string, float, boolean, and None**. Print the list. *Hint: Lists can store mixed data types.*
3. Given:

```
numbers = [10, 20, 30, 40, 50]
```

Print the element at index 2. *Hint: Indexing starts from 0.*

4. What happens if you try to access an index that does not exist? Try it and observe. *Hint: Look at the error message.*

B. Looping Through a List

1. Print all elements of a list using a **for loop with range**. *Hint: Use range(len(list)).*
2. Print all elements using a **for-each loop**. *Hint: for item in list.*
3. Print each element along with its index number. *Hint: Use range or enumerate.*
4. Count how many elements are in a list **without using len()**. *Hint: Use a counter variable.*

C. Length & Index

1. Find the length of the list:

```
cities = ["Kathmandu", "Pokhara", "Lalitpur"]
```

Hint: Use len().

2. Print total number of students using a list. *Hint: Combine len() with print().*
3. Find the index of "geeta" in:

```
dance_list = ["ram", "hari", "geeta", "shyam"]
```

Hint: Use index().

4. What happens if index() is used for an element that does not exist? *Hint: Try and observe the error.*

D. Updating List Items (Mutability)

1. Change “ram” to “Ram” in a list. *Hint: Use index assignment.*
2. Change the last element of a list to “END”. *Hint: Use len(list)-1.*
3. Take a name from user and replace it with “Student”. *Hint: Find index first.*

E. Adding Elements

1. Create an empty list and add three user-entered names. *Hint: Use append().*
2. Add an element to the end of a list. *Hint: append().*
3. Insert an element at index 0. *Hint: insert(index, value).*
4. Explain the difference between append() and insert(). *Hint: Position matters.*

F. Removing Elements

1. Remove the last element from a list. *Hint: pop().*
2. Remove element at index 1. *Hint: pop(index).*
3. Remove a specific value from a list. *Hint: remove(value).*
4. What happens if you remove an element that does not exist? *Hint: Observe the error.*
5. Remove a name only if it exists in the list. *Hint: Use “in” operator.*

G. User Input with Lists

1. Ask user for five student names and store them in a list. *Hint: Loop + append.*
2. Store user input in a list after capitalizing each name. *Hint: Use capitalize().*
3. Ask user which name to rename and update it. *Hint: index() + assignment.*
4. Ask user for a name and check if it exists in the list. *Hint: Use “in”.*

H. Logical & Thinking Questions

1. Why is a list called mutable? Show with example. *Hint: Modify after creation.*
2. Can a list store another list? Try it. *Hint: Nested list.*
3. Count how many names start with the letter “r”. *Hint: Check first character.*
4. Remove all duplicate elements from a list. *Hint: Check before adding.*
5. Print only names with length greater than 4. *Hint: Use len().*

I. Mini Challenges

1. Create a menu-driven program:

```
1. Add name
2. Remove name
3. Show list
4. Exit
```

Hint: Use while loop.

2. Take five numbers from user and print the largest one. *Hint: Compare values in loop.*
3. Reverse a list without using reverse(). *Hint: Loop from last index.*
4. Capitalize all names in a list. *Hint: Loop and update each element.*